

The emergence and cultural persistence of ineffective solutions

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Abstract

Through cumulative cultural evolution, populations discover novel solutions that allow them to solve complex problems. However, populations also reliably develop and sustain practices that are costly yet appear to have no effect on their stated outcome, such as bloodletting, divination, and malicious sorcery. Here, we show that ineffective, costly solutions can develop and be maintained because of their subjective appeal. In a pre-registered transmission-chain experiment, participants were tasked with developing a solution to a problem over several trials, after which they passed instructions onto the subsequent generation. The visual appearance of the experimental task was manipulated to vary the extent to which cognitive biases could interfere with participants' expectations of efficacy, enabling a direct test of the effect of cognitive appeal on cumulative cultural evolution. Participants in the low-appeal treatment ($n = 240$) progressively developed more efficient solutions across generations. Participants in the high-appeal treatment ($n = 240$), in contrast, produced costlier solutions with no increase in efficiency over the course of the experiment. In the high-appeal treatment, the solutions most frequently transmitted between generations were three to five times more expensive than those most frequently transmitted in the low-appeal treatment. These results suggest that, even under conditions favorable to cumulative cultural evolution, cognitive appeal alone can prevent the process and promote the emergence and cultural persistence of ineffective solutions.

Introduction

Cumulative cultural evolution appears critical for humans' success¹⁻³. Through the accumulation of cultural innovations over many generations, populations have developed solutions that no single individual could have invented, allowing us to inhabit new environments and achieve feats unparalleled in the animal world, from space exploration to industrial-scale agriculture. Several experiments have shown that cumulative cultural evolution occurs through instrumental, payoff-biased social learning, which allows the gradual accumulation of beneficial modifications over time⁴⁻⁶. Participants tasked with solving a problem, such as building a basket to carry rice or designing a wheel to roll down a track, preferentially adopt and retain solutions that yield greater expected payoffs^{7,8}. When successive generations of learners select both successful demonstrators and effective-seeming solutions, cumulative cultural evolution can drive the emergence of complex, highly-efficient solutions^{6,8}.

Theoretical and empirical work has shown that cumulative cultural evolution operates under a wide range of conditions, albeit with differing levels of effectiveness. When payoffs are immediate and accurate, payoff-biased social learning enables individuals to filter out less efficient solutions, gradually bringing populations closer to the optimal solution over time^{9,10}. However, when payoffs are noisy or delayed, gradual improvements are slower, as individuals struggle to accurately assess the effectiveness of solutions¹¹. Additionally, experimental work has demonstrated that populations tend to converge on less efficient solutions in environments that are misaligned with their prior assumptions¹². These mitigating factors notwithstanding, cumulative cultural evolution remains powerful enough to foster the emergence of increasingly efficient solutions across diverse conditions, even when those solutions are counterintuitive and beyond individual understanding⁸.

However, while cumulative cultural evolution reliably promotes the emergence of increasingly efficient solutions in experimental settings, it also appears to produce and preserve ineffective practices in the real world. Humans around the world engage in practices that seem to have no effect on their stated outcome, such as divination^{13,14}, rain magic¹⁵, malicious sorcery¹⁶, and bloodletting¹⁷. These practices are widespread and elaborate, requiring non-trivial investments of time and resources. Research at the intersection of cognitive science and anthropology suggests that these practices may develop because of systematic biases in how individuals evaluate solutions. For example, participants in the United States and Brazil evaluated rituals as more effective when they involved features like repetition or specific times,

despite the rituals having no effect on their intended outcome ¹⁸. Likewise, anthropologists have long observed that people tend to favor magic and folk medicines that rely on visual similarity between a remedy and the ailment or affected body part. A well-known example is the doctrine of signatures, which holds that herbs resembling specific body parts can be used to treat ailments affecting those parts ^{19,20}. Such common features have been hypothesized to tap into widespread cognitive biases ^{21,22}.

Here, we hypothesize that subjective appeal can drive the emergence and persistence of ineffective solutions even in conditions otherwise conducive to cumulative cultural evolution. To test this hypothesis, we conducted a pre-registered experiment in which participants were tasked with maximizing their score by allocating nutrients to nine plants in a virtual field (Figure 1A). Nutrients came in three colors of different costs—blue (2 units), yellow (6 units), and red (10 units)—and participants could allocate up to sixteen nutrients across the nine plants. This resulted in a vast solution space, with over 100 billion distinct configurations. The field's output varied as a function of the number of plants that were given nutrients, although scores were noisy through the inclusion of a random component (see Methods section). The optimal solution was to provide each plant with a single nutrient with the lowest cost (the blue nutrient).

Participants were organized into experimental chains of four generations, with each generation consisting of three participants (Figure 1B). Each participant completed six trials and received feedback on their score after each one. After their final trial, participants wrote a piece of advice for the next generation. From the second generation onward, participants were shown the final scores and written advice of three individuals from the previous generation and were asked to choose one as their cultural demonstrator before beginning their first trial.

We implemented two experimental treatments that differed in the degree of visual similarity between the features of the plants and those of the nutrients. In the low-appeal treatment (n=240), the plants had different shapes but were all gray in color (Figure 1A). In the high-appeal treatment (n=240), in contrast, plants were additionally assigned one of six colors: plant colors were either primary colors, matching the colors of the nutrients (blue, yellow, red), or secondary colors that can typically be created by mixing the primary colors (green, purple, orange). Thus, in this treatment, the nutrients fed to the plants could resemble the plants themselves, potentially allowing for more visually appealing solutions than in the low-appeal treatment. Importantly, the size of the solution space was kept equivalent across experimental treatments by pairing each plant shape with a unique color (Figure 1A).

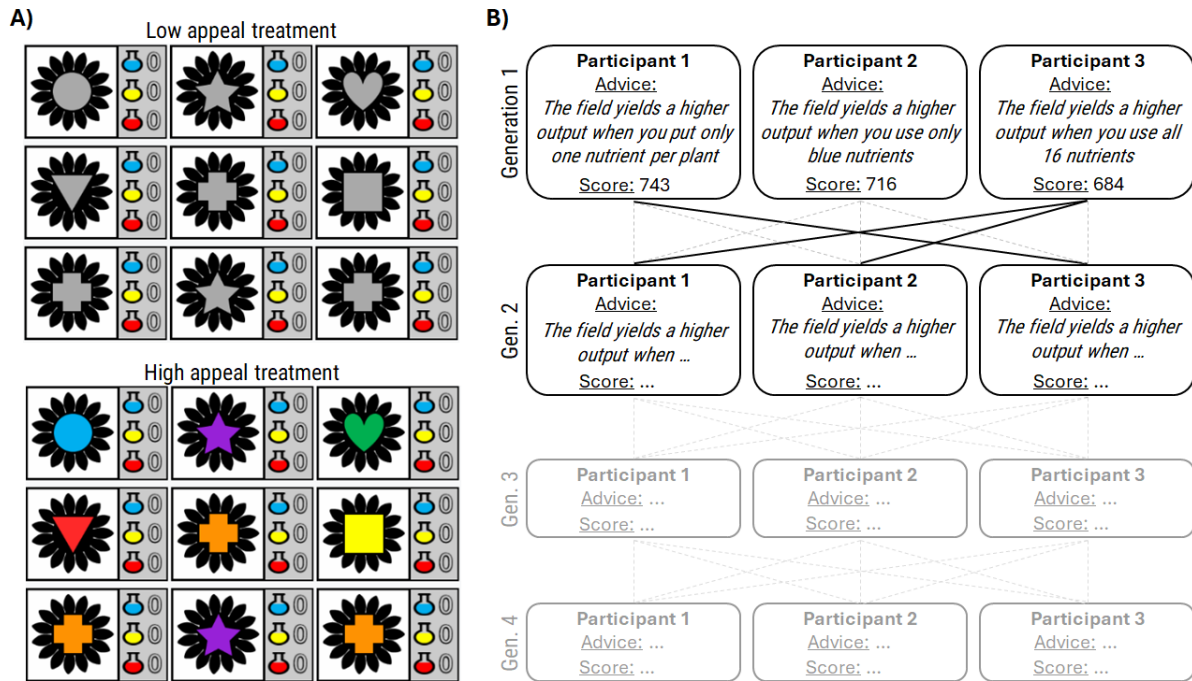


Figure 1. Problem-Solving Task and Experimental Design. A) Plant-Feeding Task: Participants were presented with six uniquely shaped flowers within a field of nine plants. The plants appeared either uniformly gray (low-appeal treatment) or in various colors, with each shape paired to a specific color (high-appeal treatment). Participants had access to three different nutrient types and could use up to 16 nutrients across the entire field. After each trial, participants received feedback on the score their solution generated. B) Participants were organized into experimental chains consisting of four generations, with three participants per generation. After engaging with the task for six trials, each participant wrote a piece of advice for participants in the subsequent generation. From Generation 2 onward, participants could select one cultural demonstrator based on that individual's written advice and score in their final trial before solving the task and producing their own piece of advice for participants in the following generation.

Under these conditions, rates of cumulative cultural evolution should be comparable across treatments if cognitive biases do not influence participants' expectations of efficacy. Conversely, if changes in the visual appearance of stimuli interfere with these expectations, we would expect rates of cumulative cultural evolution to differ between treatments. To test this, we compared the score and cost of culturally transmitted solutions—that is, the solutions produced on the sixth trial—across generations in the low-appeal and high-appeal treatments. We predicted that the score of transmitted solutions would increase over generations in the low-appeal treatment and that this increase would exceed any change observed in the high-appeal treatment ([link to the preregistration](#)).

Results

Results show that, in the low-appeal treatment, the score of the transmitted solutions increased over the four generations (Figure 2A), while the cost of these solutions progressively decreased (Figure 2B, see Methods for details on score and cost calculation). In the high-appeal treatment, in contrast, neither the score of the transmitted solutions (Figure 2A) nor the cost of these solutions (Figure 2B) changed significantly over the course of the experiment. Statistical analyses show that in the low-appeal treatment, each generation was associated with a 17-point increase in score ($p < 0.001$), whereas in the high-appeal treatment, the change in score across generations was statistically indistinguishable from zero (Supplementary Table 1A).

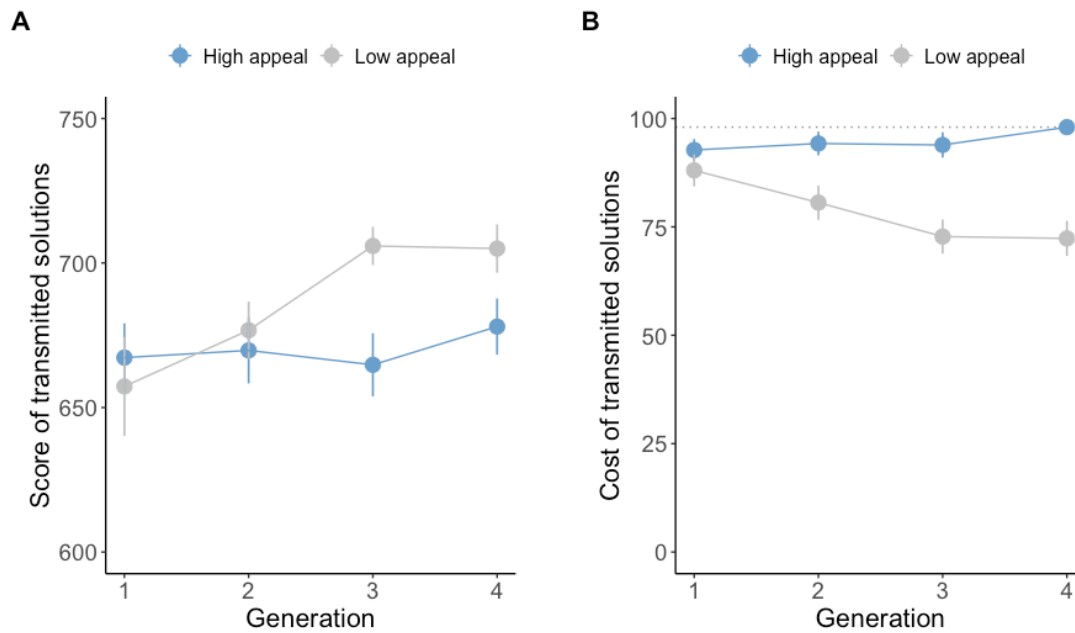


Figure 2. Score and cost of culturally transmitted solutions across generations. A) In the low-appeal treatment (gray line), participants showed gradual improvements in solution scores across generations, whereas scores in the high-appeal treatment (blue line) did not change significantly. B) Participants in the low-appeal treatment progressively reduced the cost of their solutions, while those in the high-appeal treatment showed no statistically significant change over time. The dotted line represents the cost of the mix-and-match solution (see Figure 3 for an illustration).

No differences between treatments were observed in terms of social learning strategies. Participants in both treatments relied on payoff-biased social learning to choose from the three available cultural models, selecting the highest-scoring demonstrators 72% of the time in the low-appeal treatment and 78% of the time in the high-appeal treatment ($\beta = -0.06$; 95% CI [-0.16, 0.05]; $p = 0.31$).

To understand why cumulative performance differed between treatments despite similar social learning strategies, we conducted exploratory analyses comparing exploration and cultural transmission patterns. Results from these analyses show that participants explored

the solution space more extensively in the low-appeal treatment than in the high-appeal treatment. On average, they produced 5.43 unique solutions out of 6 trials in the low-appeal treatment, compared to 4.61 in the high-appeal treatment. Across all experimental replicates, 77% of solutions were unique in the low-appeal treatment, compared to only 52% in the high-appeal treatment. Figure 3 shows the cumulative number of unique solutions across trials, illustrating that exploration was broader in the low-appeal condition.

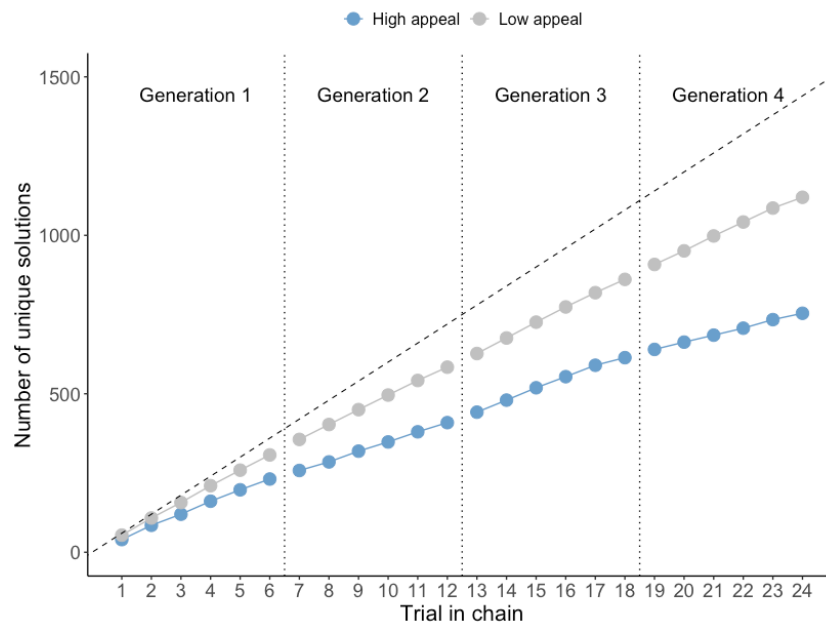


Figure 3. Exploration across trials. Cumulative number of unique solutions (i.e., nutrient-to-plant allocation patterns) across trials in each treatment. The dashed black line indicates the theoretical upper bound of exploration, assuming every participant submitted a solution not previously produced in this treatment. The gray line shows the cumulative number of unique solutions observed in the low-appeal treatment, while the blue line shows the high-appeal treatment.

Analyses of the solutions produced at the sixth trial—that is, the solutions transmitted between generations—reveal important differences between treatments. In the low-appeal treatment, the most common solution accounted for 3% of sixth-trial responses, and the second most common appeared in 2.5% of cases. Both solutions involved only blue nutrients: the most common used 16 blue nutrients, while the second most common—which was the optimal solution—used 9 blue nutrients, one per plant (Figure 4). In the high-appeal treatment, one solution accounted for 16% of sixth-trial responses. This solution followed a costly 'mix-and-match' strategy: participants matched nutrient colors to flower colors whenever possible, and when exact matches were unavailable, they combined nutrient colors based on primary-color mixing (e.g., mixing blue and red nutrients for a purple flower; Figure 4). The second most common solution, a close variation using 16 rather than 15 nutrients, appeared in 6% of cases.

These mix-and-match solutions cost approximately five times more than the optimal solution observed in the low-appeal treatment. This suggests that transmitted solutions in the low-appeal treatment were shared because they were objectively efficient, whereas in the high-appeal treatment, transmission appears to have been driven by biased expectations of efficacy rather than actual performance.

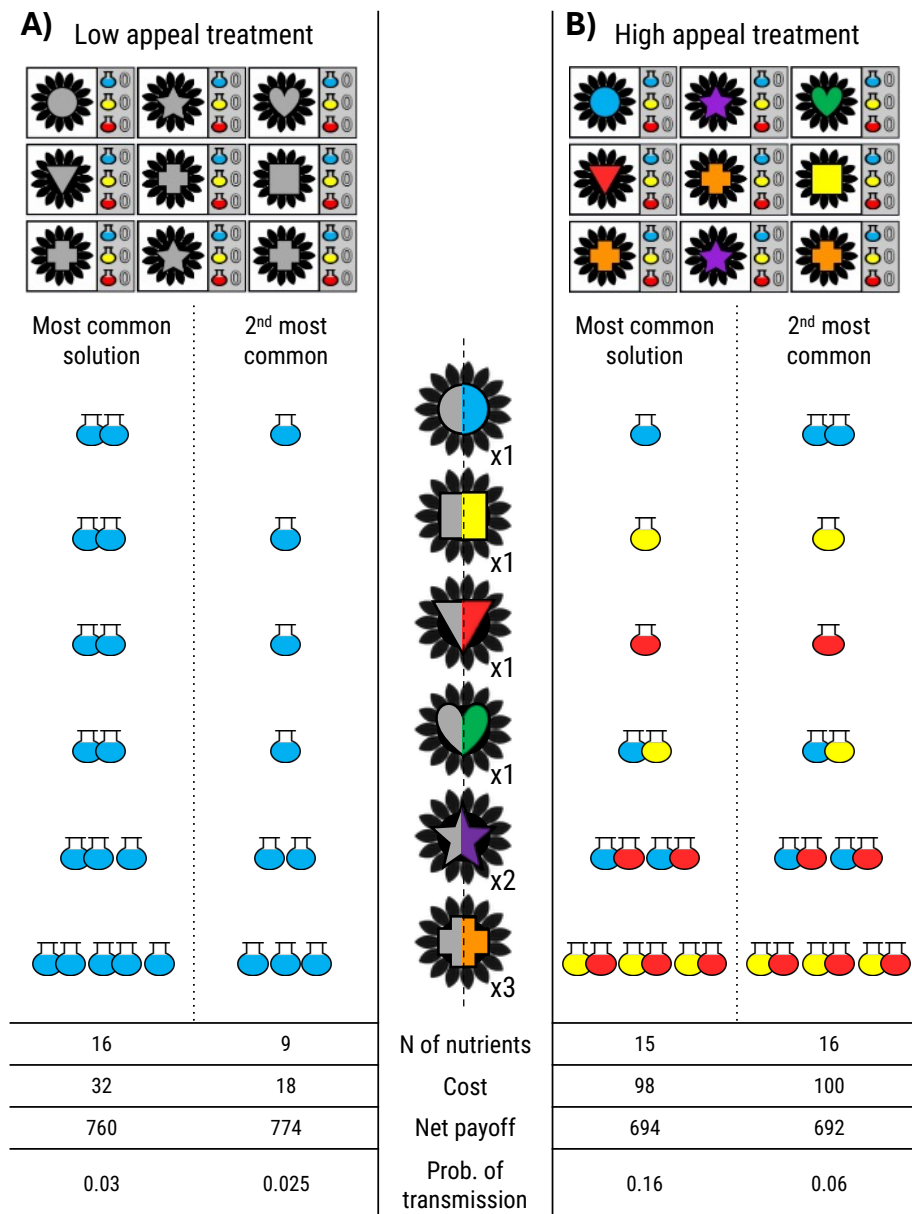


Figure 4. Most common culturally transmitted solutions. The central column shows the composition of the field, alongside the nutrient configurations used in the most common solutions for each treatment. Note that the star/violet and orange/cross plants appeared twice and three times, respectively, in the field. A) The two most common culturally transmitted solutions in the low-appeal treatment exclusively involve blue nutrients. The most common solution in this treatment uses 16 blue nutrients, the most efficient option when all nutrients are used. The second most common solution involves 9 blue

nutrients—one for each plant—which is the optimal solution. B) In the high-appeal treatment, the two most common solutions use a mix-and-match strategy, where primary-colored plants are fed with the corresponding colored nutrient, while secondary-colored plants receive a mix of colored nutrients.

Discussion

Previous research has established that payoff-biased social learning can drive the emergence of increasingly efficient solutions, as individuals tend to adopt and retain solutions that yield higher rewards⁴⁻⁶. In our study, participants similarly relied on payoff-biased social learning, most often selecting the highest-scoring cultural demonstrators. This strategy led to cumulative improvement in the low-appeal treatment, where solutions became progressively more efficient across generations. In contrast, no such improvement was observed in the high-appeal treatment: across generations, participants produced costly, ineffective solutions. These findings demonstrate that changes to the visual appearance of stimuli—by increasing the opportunity for cognitive biases to interfere with expectations of efficacy—can hinder cumulative cultural evolution, even under conditions that typically support adaptive cultural change.

Our analyses show that subjective appeal hinders cumulative cultural evolution by reducing exploration at both individual and group levels. Participants in the high-appeal treatment explored the solution space less extensively, resulting in lower cultural variation. This reduction in variation constrains cultural evolvability—the capacity of a cultural system to generate, retain, and build upon culturally transmitted variation—by limiting the raw material on which selection operates²³.

The observed decrease in exploration seems to reflect the strong appeal of the “mix-and-match” strategy in the high-appeal treatment, wherein participants aligned nutrient colors with flower colors whenever possible or combined nutrient colors based on primary-color mixing. The core feature of mix-and-match—a perceptual alignment between the intervention and the target object—mirrors patterns found in magic and folk remedies, exemplifying what anthropologists term the “law of similarity”²⁴. Such cultural patterns suggest that individuals reliably develop tendencies to associate visual similarity with functional relationships, thereby biasing their assessment of causal relationships. Research on analogical reasoning supports the idea that people often form assumptions about functions based on visual resemblance²⁵. Such intuitions can lead individuals to overestimate causal links between features and effects, such as assuming that a plant’s shape is linked to its medicinal properties or, as in our experiment,

that nutrients are more effective when their colors match those of the plants to which they are applied.

Given the apparent functionality of many cultural practices found across much of culture, researchers have often explained those that seem instrumentally ineffective, such as shamanism, magic, and divination, by proposing alternative functional benefits, such as enhanced confidence or social cohesion ^{26,27,28}. However, consistent with recent arguments ^{13,29,30}, our results experimentally confirm that such functional benefits are not necessary to explain the persistence of ineffective cultural practices. Instead, subjective appeal appears potent enough to promote the emergence and cultural persistence of these practices.

Our study has important implications for studying cumulative cultural evolution under controlled conditions. Although researchers have investigated the conditions that constrain cumulative cultural evolution ^{5,12,31-36}, most experiments are designed in ways that may artificially inflate the potential for gradual cultural improvement. For example, with few exceptions ¹¹, researchers typically provide participants with immediate and accurate payoffs, despite many interventions in the real-world leading to payoffs that are noisier and time-delayed. More importantly, the solution space in laboratory experiments is often artificially constrained, limiting the opportunity for inefficient but cognitively appealing solutions to compete with more efficient ones. These design choices potentially lead to an underappreciation of the inefficiencies of cumulative cultural evolution as manifested both in practices that are cheap and effective yet rare, such as handwashing ³⁷, and in practices that are common and costly yet futile.

In our experiment, the high-appeal treatment implemented conditions likely to engage participants' intuitive tendency to associate visual similarity with functional relationships. By embedding this cognitive bias into the task environment, we created a setting in which visually appealing—but costly and inefficient—solutions could plausibly emerge. This does not mean, however, that the experiment was artificially tailored to produce such outcomes. First, the low-appeal treatment confirms that the overall conditions of the experiment—such as task complexity, payoff noise, and number of demonstrators—were conducive to cumulative cultural evolution. Second, even in the high-appeal treatment, the opportunity for cognitively appealing solutions to emerge was limited. In real-world contexts, learners routinely encounter similarity-based cues alongside many other sources of biased inference. Outside the lab, individuals interact with a broader set of variables—such as timing and other contextual factors—that may trigger additional biases not captured in our design ¹⁸. This broader and more

complex inferential landscape may further amplify the emergence and persistence of ineffective solutions, underscoring the importance of understanding how subjective appeal shapes the dynamics of cultural evolution across contexts and domains.

Methods

Ethics Statement. All methods were approved by the Toulouse School of Economics/Institute for Advanced Studies in Toulouse Review Board for Ethical Standards in Research (2022-11-001). All participants provided informed consent before taking part in the experiment. All individual information has been anonymized.

Participants. A total of 480 participants were recruited for the study. Participants were recruited through Prolific and filtered based on several criteria: they were required to be aged between 18 and 35 years, from the United Kingdom, United States, Ireland, Australia, Canada, or New Zealand, and to have English as their first language. Additionally, participants needed to have no issues seeing colors and maintain an approval rate of 98-100% on Prolific. Participants received a £1.2 participation fee, with an average bonus of £1.8 based on their performance.

Experimental task. The experiment relied on a plant-feeding task in which participants were presented with a virtual field containing nine plants (Figure 1A). Participants aimed to maximize the output of their virtual field across 6 trials by feeding the plants with three types of nutrients that varied in their cost: blue (2 units), yellow (6 units), and red (10 units). Participants' decisions in each trial of nutrient allocation would impact the output of their field and the cost of their solution, which affected their score for that trial. The field's output was determined by the number of plants that receive at least one nutrient and a random component. Additional nutrients on the same plant, the shapes and colors of plants, or the color of the nutrients used had no effect on the output. Put precisely, the field's output was the number of plants fed, multiplied by a uniformly distributed number between 60 and 100. Subtracting the total nutrient cost and adding a fixed score of 72 resulted in the participant's score on that trial:

$$\text{Score} = 72 + \text{Uniform}(60, 100) \times \text{Number of plants fed} - \text{Total cost of nutrients}$$

The fixed score of 72 points was added to prevent participants from having a negative score in the worst-case scenario.

Participants were limited to using a maximum of 16 nutrients across the entire field, with each plant receiving no more than 9 nutrients of a given color. Under these constraints, trial scores could range from 0 to 954. The optimal strategy (feeding each plant a single blue nutrient) yielded an average score of 774. However, in the best-case scenario, the field could produce an output of 900 (9 plants $\times$ 100), at a cost of 18, resulting in a total score of $72 + 900 - 18 = 954$. In contrast, the worst strategy (feeding a single plant 9 red and 7 yellow nutrients) yielded an average score of 20. In the worst-case scenario, this approach could produce a field output of just 60 at a cost of 132, resulting in a total score of $72 + 60 - 132 = 0$.

Participants were informed about the random component affecting their score. At the end of each trial, they were shown their score for that trial. However, they were not told the exact function governing the field's output and the costs of the nutrients.

Treatments. The study implemented two treatments: low-appeal and high-appeal (see Figure 1A). A total of 480 participants were equally and randomly assigned to one of the two treatment conditions: high-appeal ($n = 240$) and low-appeal ($n = 240$). Each condition consisted of 20 transmission chains, each comprising four generations of three participants, for a total of 12 participants per chain. In the low-appeal treatment, all plants were gray and exhibited one of six distinct shapes: round, star, heart, triangle, cross, and square (see Figure 1A, upper figure). The high-appeal treatment featured the same shapes but in distinct colors: the round shape was blue, the star was purple, the heart was green, the triangle was red, the cross was orange, and the square was yellow (see Figure 1A, lower figure). All participants in each treatment saw the same fixed arrangement of the plants. In both treatments, the shapes and colors of the plants were purely aesthetic differences and did not influence the output of the field.

Cultural transmission. The participants engaged with the task over six trials. Each participant's last solution and its corresponding score for that trial were passed onto the next participant in their chain. In addition to transmitting their solutions, each participant was required to provide an accompanying piece of advice for the next participant in their chain (see Figure 1B). The advice was pre-formatted to begin with the phrase "*The field yields a higher output when...*" to ensure comparability across responses. Before starting the task, participants were provided with three pieces of advice from participants in the previous generation, along with their score from the last trial, and could select one participant as cultural demonstrator. Upon selecting a cultural demonstrator, participants were provided with the chosen piece of advice along with the participants' solution (i.e., which nutrients were provided to each plant) for the duration of the experiment.

Statistical analysis.

Score and cost of transmitted solutions. We used the transmitted solutions of each of the 480 participants as the unit of analysis in our models. The response variable was each participant's expected score, that is, the score excluding the effect of the random component, to reduce the impact of artificial noise on the statistical analyses. To compare how the score changed over the course of the experiment across the two treatment conditions, we first conducted two separate models using subsets of the data for each treatment ($n = 240$ for each, $n_{\text{chain}} = 20$). To assess how the expected score changed across generations within a given treatment condition, we used a multilevel regression model with an interaction term between generation and treatment as a fixed effect, with the pooled data (high-appeal: $n = 240$, low-appeal: $n = 240$, $n_{\text{chain}} = 40$). We repeated those three models using participants' costs as the response variable. In all models, we accounted for chain-specific random effects.

Social learning strategies. The response variable was a binary indicator of whether participants selected the highest-scoring demonstrator among the three available cultural models. We used a multilevel model with treatment as a fixed effect and included chain and generation as random effects.

All data processing was performed using the R software environment ³⁸, and we conducted multilevel regression analyses with the lmer function from the lme4 package in R ³⁹.

Pre-registration. The experiment was pre-registered on AsPredicted prior to data collection ([link](#)).

Data and Code Availability. The data and scripts for the analyses are publicly available at: <https://github.com/aseyq/inefficient-solutions>

References

- 1 Henrich, J. *The secret of our success: how culture is driving human evolution, domesticating our species, and making us smarter*. (Princeton University Press, 2015).
- 2 Boyd, R., Richerson, P. J. & Henrich, J. The cultural niche: Why social learning is essential for human adaptation. *Proc. Natl. Acad. Sci. U. S. A.* **108**, 10918-10925 (2011).
- 3 Derex, M. Human cumulative culture and the exploitation of natural phenomena. *Philosophical Transactions of the Royal Society B: Biological Sciences* **377**, 20200311 (2021). <https://doi.org/https://doi.org/10.1098/rstb.2020.0311>
- 4 Mesoudi, A. An experimental comparison of human social learning strategies: payoff-biased social learning is adaptive but underused. *Evol. Hum. Behav.* **32**, 334-342 (2011).

- 5 Derex, M., Beugin, M.-P., Godelle, B. & Raymond, M. Experimental evidence for the influence of group size on cultural complexity. *Nature* **503**, 389-391 (2013). <https://doi.org/https://doi.org/10.1038/nature12774>
- 6 Thompson, B., Opheusden, B. v., Sumers, T. & Griffiths, T. L. Complex cognitive algorithms preserved by selective social learning in experimental populations. *Science* **376**, 95-98 (2022). <https://doi.org/doi:10.1126/science.abn0915>
- 7 Zwirner, E. & Thornton, A. Cognitive requirements of cumulative culture: teaching is useful but not essential. *Scientific Reports* **5**, 16781 (2015). <https://doi.org/10.1038/srep16781>
<http://www.nature.com/articles/srep16781#supplementary-information>
- 8 Derex, M., Bonnefon, J.-F., Boyd, R. & Mesoudi, A. Causal understanding is not necessary for the improvement of culturally evolving technology. *Nature Human Behaviour* **3**, 446-452 (2019). <https://doi.org/https://doi.org/10.1038/s41562-019-0567-9>
- 9 Henrich, J. Demography and cultural evolution: How adaptive cultural processes can produce maladaptive losses - The Tasmanian case. *Am. Antiq.* **69**, 197-214 (2004).
- 10 Lehmann, L. & Wakano, J. Y. The handaxe and the microscope: individual and social learning in a multidimensional model of adaptation. *Evol. Hum. Behav.* **34**, 109-117 (2013).
- 11 Caldwell, C. A. & Millen, A. E. Conservatism in laboratory microsocieties: unpredictable payoffs accentuate group-specific traditions. *Evol. Hum. Behav.* **31**, 123-130 (2010).
- 12 Thompson, B. & Griffiths, T. L. Human biases limit cumulative innovation. *Proceedings of the Royal Society B: Biological Sciences* **288**, 20202752 (2021). <https://doi.org/doi:10.1098/rspb.2020.2752>
- 13 Hong, Z. & Henrich, J. The Cultural Evolution of Epistemic Practices. *Human Nature* **32**, 622-651 (2021). <https://doi.org/10.1007/s12110-021-09408-6>
- 14 Boyer, P. Why Divination?: Evolved Psychology and Strategic Interaction in the Production of Truth. *Curr. Anthropol.* **61**, 100-123 (2020). <https://doi.org/10.1086/706879>
- 15 Hong, Z., Slingerland, E. & Henrich, J. Magic and Empiricism in Early Chinese Rainmaking: A Cultural Evolutionary Analysis. *Curr. Anthropol.* **65**, 343-363 (2024). <https://doi.org/10.1086/729118>
- 16 Singh, M. Magic, Explanations, and Evil: The Origins and Design of Witches and Sorcerers. *Curr. Anthropol.* **62**, 2-29 (2021). <https://doi.org/10.1086/713111>
- 17 Miton, H., Claidière, N. & Mercier, H. Universal cognitive mechanisms explain the cultural success of bloodletting. *Evol. Hum. Behav.* **36**, 303-312 (2015). <https://doi.org/10.1016/j.evolhumbehav.2015.01.003>
- 18 Legare, C. H. & Souza, A. L. Evaluating ritual efficacy: Evidence from the supernatural. *Cognition* **124**, 1-15 (2012). <https://doi.org/https://doi.org/10.1016/j.cognition.2012.03.004>
- 19 Rozin, P. & Nemeroff, C. in *Heuristics and Biases: The Psychology of Intuitive Judgment* (eds Thomas Gilovich, Dale Griffin, & Daniel Kahneman) 201-216 (Cambridge University Press, 2002).
- 20 Bennett, B. C. Doctrine of Signatures: An Explanation of Medicinal Plant Discovery or Dissemination of Knowledge? *Economic Botany* **61**, 246-255 (2007).
- 21 Hong, Z. A cognitive account of manipulative sympathetic magic. *Religion, Brain & Behavior* **12**, 254-270 (2022). <https://doi.org/10.1080/2153599X.2021.2006294>

- 22 Nemeroff, C. & Rozin, P. in *Imagining the Impossible: Magical, Scientific, and Religious Thinking in Children* (eds Karl S. Rosengren, Carl N. Johnson, & Paul L. Harris) 1-34 (Cambridge University Press, 2000).
- 23 Schimmelpfennig, R., Razek, L., Schnell, E. & Muthukrishna, M. Paradox of diversity in the collective brain. *Philosophical Transactions of the Royal Society B: Biological Sciences* **377**, 20200316 (2022). <https://doi.org/doi:10.1098/rstb.2020.0316>
- 24 Frazer, J. G. & Fraser, R. *The Golden Bough: A Study in Magic and Religion*. (Oxford University Press, 1998).
- 25 Gentner, D. & Markman, A. B. Structure mapping in analogy and similarity. *American Psychologist* **52**, 45-56 (1997). <https://doi.org/10.1037/0003-066X.52.1.45>
- 26 Damisch, L., Stoberock, B. & Mussweiler, T. Keep Your Fingers Crossed! How Superstition Improves Performance. *Psychol. Sci.* **21**, 1014-1020 (2010).
- 27 Watson-Jones, R. E. & Legare, C. H. The social functions of shamanism. *Behav. Brain Sci.* **41**, e88 (2018). <https://doi.org/10.1017/S0140525X17002199>
- 28 Moore, O. K. Divination - A New Perspective. *Am. Anthropol.* **59**, 69-74 (1957).
- 29 Singh, M. Subjective selection and the evolution of complex culture. *Evolutionary Anthropology: Issues, News, and Reviews* **31**, 266-280 (2022). <https://doi.org/https://doi.org/10.1002/evan.21948>
- 30 Singh, M. The cultural evolution of shamanism. *Behav. Brain Sci.* **41**, e66 (2018). <https://doi.org/10.1017/S0140525X17001893>
- 31 Griffiths, T. L., Kalish, M. L. & Lewandowsky, S. Theoretical and empirical evidence for the impact of inductive biases on cultural evolution. *Philosophical Transactions of the Royal Society B: Biological Sciences* **363**, 3503-3514 (2008). <https://doi.org/10.1098/rstb.2008.0146>
- 32 Schillinger, K., Mesoudi, A. & Lycett, S. J. Copying error and the cultural evolution of "additive" vs. "reductive" material traditions: an experimental assessment. *Am. Antiq.* **79**, 128-143 (2014).
- 33 Schillinger, K., Mesoudi, A. & Lycett, S. J. The impact of imitative versus emulative learning mechanisms on artifactual variation: implications for the evolution of material culture. *Evol. Hum. Behav.* (2015). <https://doi.org/http://dx.doi.org/10.1016/j.evolhumbehav.2015.04.003>
- 34 Mesoudi, A. Variable Cultural Acquisition Costs Constrain Cumulative Cultural Evolution. *PLoS One* **6**, e18239 (2011).
- 35 Derex, M., Edmiston, P., Lupyan, G. & Mesoudi, A. Trade-offs, control conditions, and alternative designs in the experimental study of cultural evolution. *Proceedings of the National Academy of Sciences* **121**, e2322886121 (2024). <https://doi.org/doi:10.1073/pnas.2322886121>
- 36 Caldwell, C. A., Schillinger, K., Evans, C. L. & Hopper, L. M. End State Copying by Humans (*Homo sapiens*): Implications for a Comparative Perspective on Cumulative Culture. *J. Comp. Psychol.* **126**, 161-169 (2012).
- 37 Bloomfield, S. F. & Nath, K. J. in *Somerset, England: International Scientific Forum on Home Hygiene*.
- 38 R: A Language and Environment for Statistical Computing (R Foundation for Statistical Computing, Vienna, Austria, 2011).
- 39 Bates, D., Mächler, M., Bolker, B. & Walker, S. Fitting Linear Mixed-Effects Models Using lme4. *Journal of Statistical Software* **67**, 1 - 48 (2015). <https://doi.org/10.18637/jss.v067.i01>

Supplementary material

A) Scores			
	Low-Appeal only	High-Appeal only	Pooled
(Intercept)	643.17 *** (13.94)	663.15 *** (13.60)	663.15 *** (13.77)
Generation	17.23 *** (4.95)	2.72 (4.85)	2.72 (4.90)
Treatment Type			-19.98 (19.48)
Generation x Treatment Type			14.51 * (6.93)
B) Costs			
(Intercept)	92.17 *** (5.40)	90.85 *** (3.09)	90.85 *** (4.40)
Generation	-5.49 *** (1.50)	1.55 (1.12)	1.55 (1.32)
Treatment Type			1.32 (6.22)
Generation x Treatment Type			-7.04 *** (1.87)
Num. obs.	240	240	480
Num. groups: chain	20	20	40

Table 1. Model summary of scores (A) and costs (B) of culturally transmitted solutions across treatments and model specifications. Separate multilevel models are provided for the low-appeal condition, high-appeal condition, and pooled data to compare changes in expected scores/costs over generations, with random intercepts controlling for *chain* structure. Standard errors are shown in parentheses and statistical significance is denoted by asterisks (*: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$).